MIL-R-49024(EL)

21 January 1974

MILITARY SPECIFICATION

RADIO SET AN/GRC-103(V)2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the following units of a radio relay equipment for use in the frequency range of 395 to 705 MHz and designated Radio Set AN/GRC-103(V)2.

Receiver, Radio R-1329( )(P)/GRC-103(V)

Amplifier-Converter AM-4317( )/GRC-103(V)

Transmitter, Radio T-983( )(P)/GRC-103(V)

Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1853( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

Cable Assembly, RF CG-3444( )/U (1 ft 6 in)

Cable Assembly, RF CG-3444( )/U (3 ft 6 in)

Cable Assembly, RF CG-3443( )/U (80 ft)

Adaptor, Connector UG-1375( )/U

Cable Assembly, Special Purpose, Electrical, Branched CX-10763( )/GRC-103(V) (3 ft 6 in)

FSC 5820

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The radio set uses the following special accessories for extension of capabilities.

Extension Kit, Mast MK-1009( )/GRC-103(V)

Converter, Frequency, Electronic CV-2500( )/GRC

Cable Assembly, Special Purpose, Electrical CX-10879( )/GRC-103(V)

Dummy Load, Electrical DA-437( )/GRC-103(V)

Cable Assembly, Special Purpose, Electrical CX-10762( )/U (5 ft)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirement for (Navy)
MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-R-45292	Radio Set AN/GRC-103(V) (Units of), Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-595	Colors

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STANDARDS - (Contd)

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment: Development and Production
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
GL-SM-B-323442	Mechanical Gages for Radio Set AN/GRC-103(V)2
SC-DL-424153	Adapter, Connector UG-1375( )/U
DL-SM-B-696000	Antenna AS-1853( )/GRC-103(V)
DL-SM-B-696003	Amplifier-Converter AM-4317( )/GRC-103(V)
DL-SM-B-696006	Receiver, Radio R-1329( )(P)/GRC-103(V)
DL-SM-B-696057	Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)
SM-A-696070	Test Procedure-Mast AB-952( )/GRC-103(V)
DL-SM-B-696151	Mast AB-952( )/GRC-103(V)
DL-SM-B-696171	Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-696175	Recv-Trans, Order Wire RT-773( )/GRC-103(V)
DL-SM-B-696203	Cable Assy, R.F. CG-3443( )/U
DL-SM-B-696207	Cable Assy, R.F. CG-3444( )/U
DL-SM-B-696210	Cable Assy, S.P. Elec. CX-10763( )/GRC-103(V)
SM-A-696325	Antenna AS-1853( )/GRC-103(V)
SM-A-696394	Test Procedure - Control-Indicator
SM-A-696443	Test Specification-Amplifer-Frequency Multiplier
SM-A-696763-1	Mixer Stage, Frequency - Alignment Procedure
SM-A-698002	Test Specification - Intermediate Frequency Amplifier
SM-A-698031	Test Specification - Electrical Frequency Limiter-Discriminator
SM-A-698035	Test Specification - Video Amplifier

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DRAWINGS

ELECTRONICS COMMAND - (Contd)

SM-A-698049	Test Specification - Radio Set AN/GRC-103( )(V)2
SM-A-698050	Test Specification - Radio Set AN/GRC-103( )(V)2 (Receiver Section)
SM-A-698051	Test Specification - Radio Set AN/GRC-103( )(V)2 (Transmitter Section)
SM-A-698241	Test Specification - Pulse Form Restorer
SM-A-698317	Test Specification - Panel Monitor
SM-A-698363	Test Specification - Power Supply
SM-A-698372	Test Specification - Power Supply
SM-A-698610	Test Specification - Amplifier-Frequency Multiplier
SM-A-698628	Test Specification - Electrical Frequency Synthesizer
SM-A-698639	Alignment Procedure - Duplexer Subassembly
SM-A-698640	Alignment Procedure - Frequency Multiplier
SM-A-698641	Alignment Procedure - Band Pass Filter
SM-A-698642	Alignment Procedure - Frequency Multiplier Group
SM-A-698643	Alignment Procedure - Signal Level Control-Monitor
SM-A-698645	Alignment Procedure - Amplifier-Converter AM-4317 ( )/GRC-103(V)
SM-A-698646	Alignment Procedure - Frequency Multiplier
SM-A-698648	Alignment Procedure - Radio Frequency Amplifier
SM-A-698649	Test Procedure - AM-4321( )/GRC-103(V)
SM-A-698664	Test Specification - Amplifier Monitor
SM-A-698685	Test Specification - Distribution Box
SM-A-698695	Test Specification - Monitor Panel
SM-A-698759	Test Specification - Distribution Panel
SM-A-698776	Test Specification - Alarm Control
SM-A-698781	Alignment Procedure - Frequency Multiplier
SM-A-698789	Test Specification - Distribution Box
SM-A-698790	Test Specification - Monitor Panel
SM-A-698805	Test Specification - Telephone Signal Converter
SM-A-698806	Test Specification - Amplifier Assembly
SM-A-698807	Test Specification - RT-773( )/GRC-103(V)
SM-A-698826	Test Specification - Radio Receiver R-1329(P)( )/ GRC-103(V)
SM-A-698831	Test Specification - Transmitter T-983(P)( )/GRC-103(V)

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3. REQUIREMENTS

3.1 First Article - When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction - The equipment shall be constructed in accordance with the requirements of this specification and the drawing and data list as indicated, including the requirements for parts, materials and processes thereon.

DL-SM-B-696006	Receiver, Radio R-1329( )(P)/GRC-103(V)
DL-SM-B-696003	Amplifier-Converter AM-4317( )/GRC-103(V)
DL-SM-B-696171	Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-696057	Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)
DL-SM-B-696175	Recv Trans, Order Wire, RT-773( )/GRC-103(V)
DL-SM-B-696000	Antenna AS-1853( )/GRC-103(V)
DL-SM-B-696151	Mast AB-952( )/GRC-103(V)
DL-SM-B-696207	Cable Assy., R.F. CG-3444( )/U
DL-SM-B-696203	Cable Assy., R.F. CG-3443( )/U
DL-SM-B-696210	Cable Assy., S.P. Elec., CX-10763( )/GRC-103(V)
SC-DL-424153	Adapter, Connector UG-1375( )/U

3.3 Parts, Materials and Processes; general - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Electrical Characteristics - Radio Set AN/GRC-103(V)2 provides for multichannel, line-of-sight, two-way communication in the frequency range of 395-705 MHz (Channels 390-1010). This version is intended primarily for use as a radio link in a communication network, which may include carrier telephone and teletypewriter sets. The radio set is used in pairs; one at each end of the line-of-sight path. The radio set may be used with 6/12 or 24 channel pulse code modulation (PCM) equipment or with 4 channel frequency division modulation (FDM) when associated with appropriate interface equipment. The system shall provide channel assignments on each half and full megacycle points (1/2 MHz spacing).

3.4.1 Transmitter Section

3.4.1.1 Output Power - The normal transmitter RF output power, without the RF duplexer, shall be not less than 25 watts at any frequency in the range of 395 to 705 MHz.

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3.4.1.2 Low Power Alarm
A visual and audible alarm is provided in the transmitter. The visual alarm shall indicate any change in RF output power that is greater than 8 dB ± 2 dB below the nominal power output rating of the transmitter. The switch associated with the audible alarm is included such that: In one switch position the alarm shall sound when the visual alarms are lighted as in the case of a failure; in the second switch position, the alarm sounds when the light is not lighted. The audible alarm thus indicates a change in state of the visual signal.

3.4.1.3 Output Tube Dissipation
The transmitter output power shall not decrease below specification limits after:

a. Running the transmitter for 15 minutes with its output terminals short-circuited.

b. Running the transmitter for 15 minutes with its output terminals open-circuited.

c. Running the transmitter for 30 minutes without drive.

3.4.1.4 Frequency Setability
It shall be possible to set the transmitter to any desired frequency at 0.5 MHz intervals in the frequency range of 395 to 705 MHz, with an accuracy of better than 0.002% corresponding to 20 Hz per MHz.

3.4.1.5 Transmitter Frequency Response
The overall frequency response of the transmitter when a modulation signal to produce 100 kHz peak deviation is applied to the video input shall conform to Table I.

TABLE I

RELATIVE ATTENUATION (db) REF 10 kHz			
FREQUENCY kHz	OPTIMUM	MINIMUM	MAXIMUM
.125	1.0	0.5	1.5
.200	.5	0	1.0
.250	.2	0	.4
1.0	0	-0.2	0.2
10	0(REF)	----	---
30	0	-0.2	.2
60	.1	0	.3
120	.15	0	.5
240	.7	.5	1.3

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3.4.1.5 Transmitter Frequency Response (contd)

TABLE I (contd)

RELATIVE ATTENUATION (dB) REF 10 kHz			
FREQUENCY kHz	OPTIMUM	MINIMUM	MAXIMUM
360	1.5	.6	2.3
480	2.7	1.9	4.2
600	4.2	3.4	6.2
720	6.0	5.3	8.8
840	8.2	7.5	11.5
960	10.7	8.5	15.0

3.4.1.6 Spurious Outputs - When measured with duplexer, the power output of any harmonic shall be at least 80 dB below that of the fundamental and the power output of any non-harmonic extraneous emission of a band outside plus and minus 15 MHz of the carrier shall be at least 80 dB below that of the fundamental. All spurious outputs over the frequency range $f_o \pm 15$ MHz shall be suppressed at least 60 dB, except for the range occupied by the modulation, where the spurious outputs shall not exceed the level of the modulation envelope.

3.4.1.7 FDM Modulation Requirements - With a 1 kHz sinusoidal signal of 191 mV applied to the transmitter video input, and the input control adjusted to give a reading of 50% on the panel meter, the transmitter peak deviation shall be between 65 kHz and 97 kHz. This requirement shall be met with the meter switch in the FDM position.

3.4.2 Receiver Section - When measured with the duplexer, the following requirements shall be met.

3.4.2.1 Frequency Response Shaping - The overall frequency response of the receiver when a frequency modulated signal of constant deviation (100 kHz peak) is applied to the receiver antenna terminal and measured at the video output terminal shall conform to Table II.

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3.4.2.1 Frequency Response Shaping (contd)

TABLE II

RELATIVE ATTENUATION (dB) REF 10 kHz			
FREQUENCY kHz	OPTIMUM	MINIMUM	MAXIMUM
0.125	1.0	-0.2	2.5
0.200	0.5	-0.2	1.0
0.250	0.2	-0.2	0.4
1	0	-0.1	0.1
10	0(REF)	----	---
30	0	-0.1	0.1
60	0.15	0.05	0.25
120	0.6	0.2	0.9
180	1.3	0.8	1.8
240	2.3	1.8	2.8
360	5.1	4.4	5.8
480	9.0	8.2	10.0
600	14.0	12.5	15.5
720	20.2	18.4	22.6
840	27.5	25.3	31.0
960	36.0	33.0	40.0

3.4.2.2 Noise Figure
The noise figure of the receiver, including the losses due to the receiver protection circuit and the duplexer, shall not exceed 9 dB between 395 and 705 MHz.

3.4.2.3 Demodulation
For received signals deviated ±300 kHz, ±75 kHz and ±45 kHz, the video outputs shall be 0.265 V ± 0.03 V RMS at the VID Connector, 0.189 V ± 0.02 V RMS at the FDM Connector and -10 dBm ± 2 dB (600 OHMS) at the O W Connector, respectively.

3.4.2.4 Received Signal Monitoring
A panel meter deflection of 60% or more shall be obtained at a received signal level of -50 dBm and 10% or more at a level of -95 dBm.

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3.4.2.5 IF Bandwidth
The IF selectivity shall have a nominal 3 dB bandwidth of 750 kHz with the response down at least 60 dB at ±2.1 MHz or more from resonance.

3.4.2.6 Receiver Protection
When the transmitter has been tuned for optimum output it shall be possible to tune the receiver duplexer section through the transmitter frequency without damage to the receiver. The circuitry shall provide protection against external signals at any frequency with power level up to +47 dBm under all conditions of receiver tuning. Protection shall be provided against pulse modulated signals of any frequency of 1 KW peak power, duration 1μs, duty cycle 0.001 and a rise time of 10 ns from 10% to 90% amplitude.

3.4.2.7 Spurious Response
The ratio of desired signal response to any spurious response shall be as follows:

a. For a maximum of 6 discrete responses the ratio shall be at least 80 dB.

b. For all other responses the ratio shall be at least 100 dB.

3.4.3 Terminal Section (System Requirements)

3.4.3.1 Input Impedance
The impedance at the video input to the transmitter shall be 91 ohms ± 10% resistive to at least 480 kHz and 91 ohms ±15% to at least 800 kHz.

3.4.3.2 Output Impedance
The output impedance of the receiver shall be 51 ohms ± 10% resistive for the 24 channel PCM output to at least 800 kHz and for the regenerated 6/12 channel PCM output such that when terminated in a resistive load of 91 ohms ±5%, the waveform is as specified in 3.4.3.7.1.

NOTE: All received RF signal levels are measured at the antenna terminals of the duplexer.

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3.4.3.3 Frequency Response (PCM)
The overall frequency response between transmitter video input and receiver video output relative to the response at 10 kHz shall conform to Table III.

TABLE III

FREQUENCY kHz	REL. ATTEN. dB REFERENCE 10 kHz		
	NOMINAL	MINIMUM	MAXIMUM
.125	2.0	1.0	3.0
.200	1.0	0	2.0
.250	.4	0	.8
.500	0	-.1	+.1
1.0	0	-.1	+.1
10	0 (REF)	---	---
30	0	-.1	.2
60	.2	.1	.4
120	.7	.5	1.0
180	1.6	1.3	2.2
240	2.9	2.5	4
360	6.5	5.7	8
480	11.7	10.2	13.5
600	18.2	16.0	21.0
720	26.2	23.1	30.5
840	35.7	31.7	41.0
960	46.7	41.2	52.5

3.4.3.4 PCM Modulation Requirements
When the modulation sensitivity is adjusted to maximum, an input voltage of 1 volt peak-to-peak shall deviate the transmitter plus and minus 300 kHz peak. Also the metering circuitry shall enable the operator to adjust the peak deviation of the transmitter to plus and minus 180 kHz and plus and minus 300 kHz with an accuracy of plus 2.2 and minus 1.25 dB with inputs between 1 volt to 4 volts peak-to-peak. The transmitter shall be considered to have met this requirement if when tested in accordance with the 12 and 24 CH PCM Metering sections of the test specified in 4.19.3 it meets the specifications thereof.

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3.4.3.5 Signal to Noise (PCM)
The transmitter shall be tuned to any of its operating frequencies and shall be operated at rated RF power output. A 10 kHz sinusoidal signal shall be applied at the video input terminals of the transmitter and the transmitter shall be adjusted for a peak deviation of plus and minus 180 kHz. The receiver output, when the receiver input is coupled to the transmitter through sufficient attenuation, shall be 0.45V ± 0.15 peak-to-peak at the unregenerated output terminals. Under these conditions the ratio of average detected signal power to the average noise power at the unregenerated video output terminals of the receiver as measured with a flat weighted meter shall be at least 12 dB when the receiver RF signal is -94 dBm and at least 31 dB when the received RF signal is -74 dBm. The transmitter shall be adjusted for rated peak deviation (plus and minus 300 kHz) and the receiver unregenerated video output shall be 3/4 ± 1/4 volt peak-to-peak, the ratio shall be at least 16 dB when the received RF signal is -94 dBm and at least 35 dB when the received signal is -74 dBm.

3.4.3.6 Distortion (PCM)
The equipment shall be arranged as in 3.4.3.5 with the receiver coupled to the transmitter through sufficient attenuation so that the RF input at the receiver input terminals (including duplexer) is -54 dBm. A sinusoidal signal at any frequency within the range of 250 Hz to 240 kHz shall be applied at the video input terminals of the transmitter at a level of 2 volts peak-to-peak. The transmitter shall be adjusted for rated deviation and the receiver output shall be 3/4 plus or minus 1/4 volt peak-to-peak at the unregenerated video output terminals. Under these conditions the total harmonic distortion measured at the unregenerated video output terminals shall not exceed a level 26 dB below the fundamental.

3.4.3.7 Pulse Regeneration (PCM)
The receiver output circuits shall be capable of generating new pulses whose amplitude, timing and pulse shape are derived from the 6 channel or 12 channel signals briefly described below.

PCM CHARACTERISTICS

Number of channels	6	12
Signal Form	(Full-band, Unipolar Binary)	
Bit Rate	288	576
Pulse Width at Half Amplitude	3.472 µsec	1.736 µsec

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3.4.3.7.1 Regenerated PCM
The regeneration shall eliminate all noise peaks less than 40% of the pulse amplitude for the range of unregenerated video output levels corresponding to input deviations of ±145 kHz peak minimum to ±230 kHz peak maximum. The regenerated signal amplitude shall be -0.1 volt ±0.1 volt for the '1' condition and -2.0 volts ±0.20 volts for the '0' condition, delivered into a load of 91 ohms ±5%, with no reversal of phase from the transmitter input or unregenerated video output. Pulse width shall be full interval with 10% to 90% rise time 10 nsec minimum, 50 nsec maximum. The 50% point shall be 0 nsec minimum,80 nsec maximum after the 50% point of the rising edge of the timing signal specified in 3.4.3.7.2.

3.4.3.7.2 Timing Signal
Timing signals shall be extracted from the incoming pulse train and shall accompany the regenerated PCM to the next unit. The timing signal amplitude shall be 2 volts ±0.2V DC isolated, delivered into a load of 91 ohms ±5%. Pulse width at the 50% amplitude point shall be 80 nsec minimum, 150 nsec maximum. Rise time of the leading edge from the 10% point to the 90% point shall be 20 nsec minimum, 50 nsec maximum. Pulse rate shall be 576 kHz for both the 6 channel and 12 channel signals.

3.4.37.3 Squelch and Alarm
The squelch shall disable the regenerated PCM signal when the signal-to-noise ratio at the unregenerated video output is less than 7 dB ±2 dB when measured as in 3.4.3.5, with a reference deviation of ±180 kHz peak. The audible and visible alarm shall operate in conjunction with the squelch to indicate locally, failure of the received signal from the remote station. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver, the audible alarm can be disabled. The sensitivity of the squelch circuit shall be such that the point of operation which indicates a received signal (lamp extinguished) shall be -98 dBm ±3 dB.

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3.4.3.8 FDM Demodulation Requirements
With a signal deviated 75 kHz peak applied to the receiver antenna terminal, the level measured across the video output connector terminated in 91 ohms shall be .189 V ± .02 V RMS. The meter of the receiver shall be in the green area with the switch in the FDM position.

3.4.3.9 Duplexers
Insertion loss at the center frequency of the pass-band of the transmitter section shall not exceed 2 dB over 90% of the operating band and shall not exceed 2.5 dB over the remainder of the operating band. Insertion loss at the center frequency of the band-pass of the receiver section shall not exceed 3 dB.

3.4.3.10 Duplexer Discrimination
The discrimination of the filter for the transmitter section shall be a minimum of 25 dB at ±16 MHz and for the receiver section a minimum of 60 dB at ± 16 MHz and 100 dB at ± 50 MHz.

3.4.3.11 Voltage Standing Wave Ratio
The voltage standing wave ratio measured at the antenna terminal of the duplexer, with the transmitter terminal terminated in a 50 ohm load and the receiver connected to the receiver connection, shall not exceed 2.2:1 at any transmitted frequency within the operating frequency range as specified in 3.4.

3.4.3.12 Systems Level Stability
The transmitter and receiver shall be connected through an attenuator with sufficient attenuation such that the power input to the receiver input terminals is minus 60 dBm. With the transmitter modulated as described in 3.4.3.4. (PCM modulation requirements) and the overall system adjusted for nominal gain, the following conditions shall be met: The system gain shall not vary more than ±1 dB in any eight hour period between the 91 ohm video input and 51 ohm video output terminals of the receiver when the received signal is varied between -95 dBm and -45 dBm under any combination of temperature, humidity or line voltage variation as specified herein.

3.4.4 Order Wire

3.4.4.1 Order Wire Operation
The order wire unit shall meet the following requirements to be compatible with Multiplexers TD-202( )/G, TD-204( )/G and TD-754( )/G.

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3.4.4.1.1 Tone Generator
When the tone generator is activated, the level measured at the TRANS "A" and TRANS "B" terminals of the order units shall be -10 dBm ± 1 dBm.

3.4.4.1.2 Signal Receiver Detector Sensitivity
When a -10 dBm ±6 dB, 1600 Hz ±4 Hz signal is applied to either the REC A or REC B terminals of the order wire unit, the signal lamp shall light and the buzzer sound.

3.4.4.2 Receiver Demodulation Order Wire
With an RF signal deviated ±45 kHz peak applied to the receiver antenna terminal, the level measured at the receiver order wire shall be -10 dBm ±2 dB.

3.4.4.3 Transmitter Deviation Order Wire
A 1000 Hz sinusoidal signal at a level of zero dBm applied to the microphone input terminals of the order wire unit shall produce a peak deviation of the transmitter between 38 kHz and 59 kHz. A 1000 Hz signal of -10 dBm applied to the transmitter order wire connector shall produce a peak deviation of between 39 kHz and 58 kHz.

3.4.4.4 Signal-To-Noise (Order Wire)
The transmitter and receiver shall be connected together in a systems arrangement with sufficient attenuation between them such that the power measured at the input terminals of the receiver is -94 dBm. With the transmitter modulated as in 3.4.4.3 the signal-to-noise ratio measured with F1A weighting at the local order wire terminals (receive phone terminals) shall be at least 29 dB at any operating frequency in the presence of a normal 6/12 channel PCM signal and at least 40 dB in the FDM mode of operation. When measured with C-message weighting the signal-to-noise ratio shall be at least 27.5 dB and 38.5 dB, respectively.

3.4.4.5 Peak Limiting (Order Wire)
The limiting action to be provided in the transmitter for the order wire shall be such as to compress test tone power inserted at the microphone terminals in accordance with Table IV.

TABLE IV

Limiting Characteristic, Insert Path

Measured between MIKE terminal and TRAN 'A' or TRAN 'B' terminals.

INPUT POWER	COMPRESSION
-10 dBm	None
-5 dBm	None
0 dBm	Less than 0.5 dB
+5 dBm	3.5 dB ± 1 dB
+10 dBm	7.5 dB ± 2 dB
+15 dBm	10.0 dB ± 3 dB

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3.4.4.6 Frequency Response (Order Wire System)
The overall frequency response characteristic between the local microphone input and the phone output terminals of the order wire unit at a distant station shall be plus 1 to minus 3 dB from 300 to 1800 Hz, referred to the 1 kHz response at an RMS level of zero dBm, when the equipment is employed in a full duplex PCM system.

3.4.4.7 Frequency Response (Order Wire Unit)
The order wire unit RT-773( )/GRC-103 shall meet the following frequency response characteristics (Table V) when measured with input levels specified in 3.4.4.11.

TABLE V

OW FREQUENCY RESPONSE

PATH	FROM	TO	FREQUENCY RANGE	RESPONSE REF: 1 kHz
'THRU'	REC 'A'	TRAN 'B'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
'THRU'	REC 'B'	TRAN 'A'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
DROP	REC'A'	PHONE terminal	200 Hz -2.0 kHz	+0.5 dB -1.5 dB
DROP	REC 'B'	PHONE terminal	200 Hz -2.0 kHz	+0.5 dB -1.5 dB
INSERT	MIKE terminal	TRAN 'A'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
INSERT	MIKE terminal	TRAN 'B'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
SIDETONE	MIKE terminal	PHONE terminal	200 Hz -2.0 kHz	±1.0 dB

3.4.4.8 Distortion (Order Wire System)
With the equipment arranged as in 3.4.4.4, Signal-to-Noise, the total harmonic distortion measured at the order wire output terminals shall not exceed 5%.

3.4.4.9 Distortion (Order Wire Units)
The distortion of the order wire unit when measured with a 1000 Hz signal as specified in 3.4.4.7 shall be as follows:

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3.4.4.9 (contd)

'THRU' PATHS	REC 'A' to TRAN 'B'	Less than 1%
	REC 'B' to TRAN 'A'	Less than 1%
INSERT PATHS	MIKE terminal to TRAN 'A'	Less than 2%
	MIKE terminal to TRAN 'B'	Less than 2%
DROP PATHS	REC 'A' to PHONE terminal	Less than 1%
	REC 'B' to PHONE terminal	Less than 1%

3.4.4.10 Net Loss (Order Wire System)
The net loss between the local microphone input and the phone output at a distant station shall be 18 dB ± 3 dB.

3.4.4.11 Order Wire Unit Levels
The order wire unit levels when measured with a 1000 Hz signal shall conform to Table VI:

TABLE VI

FROM	TO	OUTPUT LEVEL	INPUT LEVELS
REC 'A'	TRAN 'B'	-10 dBm ± 0.1 dB(adjustable)	-10 dBm
REC 'B'	TRAN 'A'	-10 dBm ± 0.1 dB(adjustable)	-10 dBm
REC'A	PHONE terminal	-18 dBm ± 1.0 dB	-10 dBm
REC 'B'	PHONE terminal	-18 dBm ± 1.0 dB	-10 dBm
MIKE terminal	TRAN 'A'	-10 dBm ± 1.0 dB	0 dBm
MIKE terminal	TRAN 'B'	-10 dBm ± 1.0 dB	0 dBm
MIKE terminal	PHONE terminal	-24 dBm ± 2.0 dB(sidetone)	0 dBm
REC 'A'	TRAN 'A'	-55 dBm ± 10.0 dB(crosstalk)	-10 dBm
REC 'B'	TRAN 'B'	-55 dBm ± 10.0 dB(crosstalk)	-10 dBm

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3.4.5 Antenna

3.4.5.1 Symmetry
The ratio of the half-power (-3dB) E- and H- plane beamwidth shall not exceed 2.0:1 over the frequency range.

3.4.5.2 Squint
The gain of the antenna measured in the direction of the normal from the antenna shall not differ from the maximum gain measured at any frequency in the operating range by more than 0.5 dB.

3.4.5.3 Minor-Lobes
The ratio of the major lobe of the antenna to any minor lobe shall be at least 12 dB at any frequency in the operating range.

3.4.5.4 Gain
The gain of the antenna relative to an isotropic radiator shall be at least 12.1 dB at 395 MHz, 12.4 dB at 450 MHz, 12.6 dB at 500 MHz, 12.8 dB at 550 MHz, 13.0 dB at 600 MHz, 13.2 at 650 MHz and 13.5 dB at 705 MHz.

3.4.5.5 Voltage Standing Wave Ratio
The VSWR measured at the antenna terminals shall not exceed 1.75:1 over the operating frequency range.

3.4.5.6 Impedance
The nominal input impedance of each antenna shall be 50.0 ohms.

3.5 Mast

3.5.1 Erection
The mast shall be capable of erection with the antenna and transmission line installed in winds of twenty-five miles per hour with the launcher guyed and all other guy wires slack.

3.5.2 Operation
The mast assembly, erected with guy wires tensioned and antenna and transmission line installed, shall withstand the following conditions without permanent deformation and maintain the antenna azimuth ±5° of any initial setting.

a. Wind velocities up to 75 mph or;

b. Wind velocities up to 50 mph with a one-half inch radial ice load on the entire mast-antenna installation.

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3.6 Finish, Protective
Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. Unless otherwise specified, the final paint film on type I surface (see MIL-F-14072), exposed to view during operation, shall be a smooth, semigloss (olive drab) green, in color conforming to No. X24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the contracting officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact. The top coat of paint on the front panel of the equipment shall be semi-gloss gray enamel in accordance with MIL-E-15090.

3.7 Marking

3.7.1 General
Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.7.2 Visibility
Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.7.3 Chassis Marking
All chassis identification, parts identification, warning notes, etc., shall be applied by the silk screen process or processes giving the same degree of legibility, placement accuracy and permanence. The marking shall be in accordance with the drawings.

3.7.4 Serial Numbers
The following items shall have serial numbers and name plates:

Transmitter, Radio T-983( )P/GRC-103(V)

Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)

Receiver, Radio R-1329( )P/GRC-103(V)

Amplifier-Converter AM-4317( ) /GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1853( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

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3.7.4.1 Composition of Serial Numbers
The serial numbers used for this equipment shall be ten digits. The first five digits shall be the Federal identification number of the prime manufacturer. The sixth digit shall be a code letter to indicate the production contract with said manufacturer. A for first production, B for second production, etc. The final four digits shall be progressively from 0000 to 9999 as required. For substantially continuous production on more than one contract consecutive numbering without change of letter designator may be used upon application to and approval by the Contracting Officer. Records will be maintained and furnished the Government of the first and last serial number of each unit, assembly, subassembly and module furnished on the contract.

3.8 Power Requirements
The equipment shall operate from a 115 volt RMS, 47.5 hertz to 420 hertz, single phase source, having a voltage variation of plus or minus 10%. The total power drain for a transmitter and receiver shall not exceed 400 watts at normal operating conditions.

3.9 Electrical Interchangeability
Like units, assemblies, subassemblies and replaceable parts shall be functionally interchangeable without modification or adjustment except operator adjustments of such items or of the equipment. Units, assemblies, subassemblies and replaceable parts shall meet the electrical requirements of their respective drawings and the tests of 4.19.

3.10 Mechanical Interchangeability
Like units, assemblies, subassemblies and replaceable parts shall be physically interchangeable without modification or adjustment except operator adjustments of such items or of the equipment.

3.11 Service Conditions

3.11.1 Temperature

3.11.1.1 Low
During testing as specified in 4.7.4.1, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.11.1.2 High
During testing as specified in 4.7.4.2, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

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3.11.2 Moisture Resistance (Humidity)
During testing as specified in 4.7.5, the equipment shall meet the performance requirements of 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1. Upon completion of tests there shall be no evidence of corrosion or degradation of finishes.

3.11.3 Elevation (Altitude)
The equipment shall meet full specification performance of 4.9.2 during and after testing as specified in 4.7.6.

3.11.4 Vibration, Bounce and Shock
The equipment shall be capable of meeting the requirements of Table VII.

TABLE VII

BOUNCE AND SHOCK

Paragraph Heading	Test Para.	Performance After Test
Bounce	4.7.1	Specified performance. (Notes A and B). No physical damage.
Shock; ballistic	4.7.2	Operable (Note B). Any physical damage shall be minor only.
Shock; bench-handling	4.7.3	Specified performance (Note B). No physical damage.

NOTE A: With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.

NOTE B: The equipment shall meet specified performance for the following:

a. Paragraph 1.3 and 1.4 of SM-A-698051, except test channel shall be 390, 759, 760 and 1010 at 115 volts, 60 Hz.

b. Paragraph 1.4 of SM-A-698050 except test channels shall be 400, 700 and 1000.

c. Paragraph 4.2.2 of SM-A-698049.

d. Paragraph 4.1 of SM-A-698049 except test channels shall be 390, 759, 760 and 1010.

e. Paragraph 4.5.2 of SM-A-698049.

f. Paragraph 4.10 of this specification.

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3.12 Electromagnetic Interference Characteristics - The equipment shall meet the electromagnetic interference requirements of MIL-STD-461, Notice 4, Table A-1 as defined herein. The following Class CE requirements apply to the AN/GRC-103(V)2:

CE06 key-up only	CS01 CS06	RE02 RE02.1	RS03.2	CE04B CS02B

3.13 Preconditioning - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the preconditioning of 4.6.

3.14 Workmanship - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

3.15 Reliability

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3.15.1 Numerical Reliability Requirement - The equipment shall have a mean-time-between-failure (MTBF) of 1200 hours when operated in accordance with Appendix I to this specification as follows:

a. Temperature: 40° ± 5°C

b. Temperature Cycling: None

c. Vibration: 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment. The duration of vibration shall be for two minutes of every operating hour during one eight-hour period each week.

d. Equipment On-Off Cycling: Equipment shall be operated continuously for periods of 144 hours and shall be switched off for periods of eight hours.

e. Input Voltage: Nominal Specified Voltage ± 5%.

f. Input Voltage Cycling: None.

3.15.2 Reliability Program - The contractor shall conduct a reliability program that shall provide for, but shall not be limited to, the following reliability tasks as described in MIL-STD-785:

a. Reliability Management (5.1)

b. Parts Reliability (5.2.3)

c. Reliability Demonstration (5.3.3)

d. Failure Data (5.4)

3.16 Technical Literature, Tools and Running Spare Parts
Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include preparation for delivery.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.24).

4.3 First Article - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units - The contractor shall furnish seven (7) first article units of the complete Radio Set AN/GRC-103(V)2.

4.3.2 First Article Inspection - The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, Group C and Group D (see Table VIII, IX, X and XI respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units; and designated units for specific subgroups within Group C in accordance with 4.5.3.1 and Group D.

4.3.3 First Article Data - The first article test plan and test report(s) shall be as required in the contract or purchase order.

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4.4 Inspection covered by subsidiary documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

ITEM	WHERE REQUIRED
Parts, Materials and Processes	3.3
Finish	3.6
Marking	3.7
Workmanship	3.14

4.5 Quality conformance inspection of equipment before preparation for delivery - The contractor shall perform the inspections specified in 4.4, 4.6, 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VIII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in Table VIII.

TABLE VIII - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(a) Visual and Mechanical	3.2 3.3, 3.6 3.7 3.14	4.20	1%	4%
Receiver, Radio (1RE1) R-1329(p)/GRC-103(V)				
Synthesizer (1RE1A2)			1%	4%
Panel, Monitor (1RE1A1)			1%	4%
Power Supply (1RE1PS1)			1%	4%
Restorer, Pulse Form (1RE1A3)			1%	4%
Limiter-Discriminator (1RE1A4)			1%	4%
Amplifier-Frequency Multiplier (1RE1A5)			1%	4%

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(a) Visual and Mechanical				
Receiver, Radio (1RE1) R-1329 (P)/GRC-103(V) - (Contd)				
Amplifier, Video (1RE1AR1)			1%	4%
Amplifier, Intermediate Freq.(1RE1AR2)			1%	4%
Case, Receiver (1A2) CY-4638/GRC-103(V)			1%	4%
Distribution Box (1A2A1)			1%	4%
Amplifier-Converter (33A1) AM-4317( )/GRC-103(V)	3.2 3.3, 3.6 3.7, 3.14	4.20	1%	4%
Duplexer (33A1A1)			1%	4%
Converter, Frequency, Electronic (33A2)			1%	4%
Amplifier, Radio Frequency (33A7)			1%	4%
Control-Indicator (33A4)			1%	4%
Control-Monitor, Signal Level (33A5)			1%	4%
Transmitter, Radio (5TR1) T-983 (P) ( )/GRC-103(V)	3.2 3.3, 3.6 3.7, 3.14	4.20	1%	4%
Synthesizer (5TR1A2)			1%	4%
Panel, Monitor (5TR1A1)			1%	4%
Power Supply (5TR1PS1)			1%	4%
Control, Alarm (5TR1A3)			1%	4%
Amplifier-Frequency Multiplier (5TR1A4)			1%	4%
Amplifier-Monitor (5TR1A5)			1%	4%
Case, Transmitter (5A2) CY-4637( )/GRC-103(V)			1%	4%
Distribution Box (5A2A1)			1%	4%
Amplifier-Frequency Multiplier (37) AM-4321( )/GRC-103(V)	3.2 3.3, 3.6 3.7, 3.14	4.20	1%	4%
Amplifier, Radio Frequency (37AR1)			1%	4%
Frequency Multiplier Assy (37A1)			1%	4%
Control-Indicator (37A2)			1%	4%

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TABLE VIII - GROUP A INSPECTION (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Receiver-Transmitter, Order Wire (9) RT-773( )/GRC-103(V)	3.2, 3.3 3.6, 3.7, 3.14	4.20	1%	4%
Panel, Distribution (9A1)			1%	4%
Panel, Monitor (9A2)			1%	4%
Converter, Telephone Signal (9A3)			1%	4%
Amplifier Assembly (9A4)			1%	4%
Case, Recvr.,-Trans., Order Wire (9MP1)			1%	4%
Antenna (21) AS-1853( )/GRC-103(V)	3.2, 3.3 3.6, 3.7, 3.14	4.20	1%	4%
Reflector, Antenna (21A1)			1%	4%
Antenna Element (21A2)			1%	4%
Mast (12) Mast AB-952( )/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Elevator, Antenna			1%	4%
Mast Section			1%	4%
Accessory Kit, Mast			1%	4%
Bag, Mast Accessories			1%	4%
Guy			1%	4%
Ring, Guy Attachment			1%	4%
Spike, Common			1%	4%
Adapter, Antenna			1%	4%
Anchor, Guy			1%	4%
Hammer			1%	4%
Strap Wrench			1%	4%
Cable Assemblies	3.2, 3.3 3.6, 3.7, 3.14	4.20		
CG-3443/U			1%	4%
CG-3444/U			1%	4%
CX-10763			1%	4%

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(b) Electrical (Subassemblies)				
Receiver, Radio R-1329( )(P)/GRC-103(V)	3.9	4.19.1		
Dimmer Switch				
Wideband Noise				
Frequency Deviation				
FDM Output			1% FOR THE GROUP	
OW Output				
PCM Regeneration				
PCM Squelch				
PCM Mute				
Receiver Squelch				
Receiver Signal Monitor				
FDM Return Loss				
Video Return Loss				
Frequency Generation				
Power Output				
Video Frequency Response				
Amplifier-Converter AM-4317/GRC-103(V)	3.9	4.19.2		
AGC			1% FOR THE GROUP	
Noise Figure				
Input VSWR				
Transmitter, Radio T-983( )(P)/GRC-103(V)	3.9	4.19.3		
Voltage Stability				
Video Input and Deviation Metering			1% FOR THE GROUP	
Power Output				
Overheat Alarm				
Low Power				
Output Frequency				
Amplitude-Frequency Response				
Video Input Match				
Amplifier Frequency Multiplier AM-4321/GRC-103(V)	3.9	4.19.4		
Initial Power Output			1% FOR THE GROUP	
Voltage Check				
Power Output and Tracking				

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(b) Electrical Subassemblies				
Receiver, Radio (1RE1) (Contd)				
Synthesizer (1RE1A2)	3.9	4.19.5		
Frequency Check Synch. Alarm Deviation Power Output			1% FOR THE GROUP	
Panel, Monitor (1RE1A1)				
Continuity Check	3.9	4.19.8	1%	
Power Supply (1RE1PS1)	3.9	4.19.9		
Initial Calibration Meter Check Regulation Load & Regulation Unregulated Supply Ripple Measurement (Unreg.) Ripple Measurement (Reg.) High Frq. Instability			1% FOR THE GROUP	
Restorer, Pulse Form (1RE1A3)	3.9	4.19.6		
Timing Pulse PCM Regenerator OW Noise Test OW Distortion			1% FOR THE GROUP	
Limiter-Discriminator (1RE1A4)	3.9	4.19.7		
Demod. Linearity Deviation Sensitivity Frequency Response			1% FOR THE GROUP	
Amplifier-Frequency Multiplier (1RE1A5)	3.9	4.19.10		
Harmonic Level			1%	

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TABLE VIII - GROUP A INSPECTION (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(b) Electrical (Subassemblies)				
Receiver Radio (1RE1) (Contd)				
Amplifier, Video (1RE1AR1)	3.9	4.19.11		
Gain				
24CH PCM			1%	
Low Signal Alarm			FOR	
Frequency Reponse			THE	
Distortion			GROUP	
Continuity Check				
Amplifier, Intermediate Frequency (1RE1AR2)	3.9	4.19.12		
Gain (without AGC)			1%	
Frequency Response			FOR THE	
AGC Limited Output Level			GROUP	
Case, Receiver (1A2) CY-4638( )/GRC-103(V)				
Distribution Box (1A2)	3.9	4.19.13		
Continuity Check				
Amplifier-Converter (33) AM-4317( )/GRC-103(V)				
Tuner, Radio Frequency (33A1)	3.9	4.19.14	1%	
Duplexer (33A1A1)	3.9	4.19.15		
Adjustments of Lead Lengths			1%	
Receiver Channel Calibration			FOR THE	
Transmitter Channel Calibration			GROUP	
VSWR Bandwidth				
Receiver Path Insertion Loss			1%	
Receiver Path Ripple			FOR	
Transmitter Path Insertion Loss			THE	
Transmitter Path Ripple			GROUP	
Local Oscillator Signal Tracking				
Converter, Frequency Electronic (33A2)	3.9	4.19.16		
Frequency Tracking			1%	
VSWR			FOR THE	
Output Power			GROUP	

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Mixer, Stage, Frequency (33A7)	3.9	4.19.17		
Gain and Bandwidth VSWR Noise Figure			1% for the group	
Control-Indicator (33A4) Coding Check	3.9	4.19.18		
Control Monitor, Signal Level (33A5)	3.9	4.19.19		
VSWR Insertion Loss Isolation Measurement			1% for the group	
Transmitter, Radio (5TR1) Panel, Monitor (5TR1A1) Continuity Check	3.9	4.19.20		
Synthesizer (5TR1A2)	3.9	4.19.5		
Frequency Check Synch. Alarm Deviation Power Output			1% for the group	
Control, Alarm (5TR1A3)	3.9	4.19.21		
Low Power Alarm and Backlash Adjustment				
Amplifier-Frequency Multiplier (5TR1A4)	3.9	4.19.22		
Frequency Response Output Level Power Output (with ALC) Harmonic Suppression Input VSWR			1% for the group	

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL * Major	AQL * Minor
Amplifier-Monitor (5TR1A5)	3.9	4.19.23		
Operational Check Metering Check Frequency Response			1% for the group	
Power Supply (5TR1PS1)	3.9	4.19.24		
Insulation and Resistance Preliminary Check Meter Reading Input Rectification High Tension Rise Time and Frequency Measurement at 5KHZ Converter Rise Time and Frequency Measurement at 400 Hz Converter 26V dc Unreg. Supply Filament Supply 28V dc/12V dc Reg. Supply High Frequency Instability Constant Current High Tension Metering Circuit Calibration 5kHz Transistor Saturation Blower Voltage and Alarm Supply Hum Measurement (120 Hz) Spike Measurement (5kHz) Input AC Current Short Circuit Protection Check Starting Circuit Check			1% for the group	
Case, Transmitter (5A2) CY-4637( )/GRC-103(V)				
Distribution Box (5A2A1) Continuity Check	3.9	4.19.25	1%	

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL * Major	AQL * Minor
Amplifier-Frequency Multiplier (37) AM-4321( )/GRC-103(V)				
Amplifier, Radio Frequency (37AR1)	3.9	4.19.26		
Drive Power Calibration				
Output Amplifier-Power Output				
Output Amplifier-Input Matching				
Driver Amplifier-Power Output			1% for the group	
Driver Amplifier-Input Matching				
Complete Amplifier-Tuning and Power Output				
Frequency Multiplier Assembly (37A1)	3.9	4.19.27		
Tuning Procedure Band 1B			1%	
Tuning Procedure Band 1A				
Control-Indicator (37A2)	3.9	4.19.28		
Coding Check			1%	
Receiver-Transmitter Order Wire (9) RT-773( )/GRC-103(V)				
Panel, Distribution (9A1)	3.9	4.19.29		
Point-to-point Resistance			1%	
Panel, Monitor (9A2)	3.9	4.19.30		
Point-to-point Resistance			1%	
Converter, Telephone Signal (9A3)	3.9	4.19.31		
Oscillator Frequency			1% for the group	
Signal Receiver Selectivity				
Signal Receiver Sensitivity				

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Amplifier Assembly (9A4)	3.9	4.19.32		
Thru Paths				
Power Gain			1%	
Frequency Response			FOR	
Harmonic Distortion			THE	
Crosstalk			GROUP	
Insert Paths				
Power Gain			1%	
Frequency Response			FOR	
Harmonic Distortion			THE	
Signal Compression			GROUP	
Drop Paths				
Power Gain			1%	
Frequency Response			FOR	
Harmonic Distortion			THE	
Sidetone Level			GROUP	
(c) Electrical (Main Assemblies)				
Receiver, Radio R-1329(P)( )/GRC-103(V) with Ampl-Conv AM-4317( )/GRC-103(V)				
	3.4.2.3			
Demodulation	3.4.3.8	4.15		
	3.4.4.2		1%	
Noise Figure	3.4.2.2	4.15	FOR	
IF Bandwidth	3.4.2.5	4.19.1	THE	
Received Signal Monitoring	3.4.2.4	4.15	GROUP	
Receiver Protection	3.4.2.6	4.18		
Frequency Response	3.4.2.1	4.15		
VSWR (Duplexers)	3.4.3.11	4.19.15		
Transmitter, Radio T-983(P)( )/GRC-103(V) with Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)				

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TABLE VIII - GROUP A INSPECTION - (Contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Frequency Response	3.4.1.5	4.16	1%	
Power Output	3.4.1.1	4.16	FOR	
Power Dissipation	3.4.1.3	4.16	THE	
Setability	3.4.1.4	4.16	GROUP	
Low Power Alarm	3.4.1.2	4.16		
Receiver-Transmitter, Order Wire (RT-773( )/GRC-103(V))				
Tone Generator	3.4.4.1.1	4.17	1%	
Deviation	3.4.4.3	4.19.3	FOR	
Sidetone	3.4.4.11	4.19.32	THE	
Peak Limiting	3.4.4.5	4.19.32	GROUP	
Antenna (AS-1853( )/GRC-103(V))				
Symmetry	3.4.5.1	4.13	1%	
Squint	3.4.5.2	4.13	FOR	
Minor Lobes	3.4.5.3	4.13	THE	
Gain	3.4.5.4	4.13	GROUP	
VSWR	3.4.5.5	4.13		
Impedance	3.4.5.6	4.13		

4.5.1.1 Order of Inspection Within Group A - Group A Inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B Inspection - Group B Inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A Inspection. This inspection shall conform to Table IX and to the special inspection levels of Table I of MIL-STD-105.

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TABLE IX - GROUP B INSPECTION

Inspection	Req Para	Insp Para	AQL Major
(a) Electrical (Main Assemblies) Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4317( )/GRC-103(V)			
Output Impedance	3.4.3.2	4.19.1	6.5%
Squelch and Alarm	3.4.3.7.3	4.19.1	6.5%
Transmitter, Radio T-983( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4321( )/GRC-103(V)			
Imput Impedance	3.4.3.1	4.19.3	6.5%
PCM Modulation	3.4.3.4	4.19.3	6.5%
(b) Electrical (System) Radio Set AN/GRC-103( )/(V)2			
Frequency Response	3.4.3.3	4.18	
PCM Signal-to-Noise	3.4.3.5	4.18	
PCM Distortion	3.4.3.6	4.18	
Pulse Regeneration	3.4.3.7.1	4.18	
Timing Signal	3.4.3.7.2	4.18	
Order Wire Distortion	3.4.4.8	4.18	
Order Wire Signal-to-Noise	3.4.4.4	4.18	
Power Requirement	3.8	4.18,4.11	
Order Wire Frequency Response	3.4.4.6	4.22	
Order Wire Net Loss	3.4.4.10	4.22	
Duplexer	3.4.3.9	4.19.15	
Duplexer Discrimination	3.4.3.10	4.19.15	
(c) Mechanical Requirements			
Mechanical Interchangeability	3.10	4.21	

4.5.2.1 Group B Sampling Plans - The Group B AQL shall be as shown in Table III and the inspection level shall be S-3.

4.5.2.2 Order of Inspection Within Group B - Group B Inspection shall be performed in the order listed in Table IX.

4.5.3 Group C Inspection - Group C Inspection shall be performed on units that have passed Group A and Group B Inspection. The inspection shall consist of the inspections specified in Table X. Samples shall be selected in accordance with 4.5.3.1.

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TABLE X - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Sub-Group I		
Mast Erection	3.5.1	4.14
Mast Operation	3.5.2	4.14
Transmitter Modulation	3.4.1.7	4.19.3
Duplexer Insertion Loss	3.4.3.9	4.19.15
Systems Level Stability	3.4.3.12	4.9.1
Shock, Bench Handling	3.11.4	4.7.3
Shock test; Ballistic	3.11.4	4.7.2
Bounce Test	3.11.4	4.7.1
Sub-Group II		
Low Temperature	3.11.1.1	4.7.4.1
High Temperature	3.11.1.2	4.7.4.2
Altitude	3.11.3	4.7.6
Humidity	3.11.2	4.7.5
Sub-Group III		
Electromagnetic Interference Test	3.4.1.6 3.4.2.7, 3.12	4.23

4.5.3.1 Sampling for Group C Inspection

4.5.3.1.1 Sub-Group I - For Sub-Group 1, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random from each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group I in the order specified.

4.5.3.1.2 Sub-Group II - For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random for each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group II in the order specified.

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4.5.3.1.3 Sub-Group III - For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 500 units. For subsequent Group C inspection, one unit shall be selected at random for each 500 units or fraction thereof. This unit shall be subjected to the Electromagnetic Interference test.

4.5.3.2 Order of Inspection Within Group C - Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Non-compliance - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of the failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative (QAR). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inpsections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group C Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected and pass Group A inspection only.

4.5.5 Group D inspection - This inspection shall consist of the test specified in Table XI and shall be performed on units from lots which have been subjected to and met Group A and B inspection.

TABLE XI - GROUP D INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph
Reliability	3.15	4.12

4.5.5.1 Sampling for Group D Inspection - Samples shall be selected at random in accordance with paragraph 4.5.5.

4.5.5.2 Group D Failures - Actions required relative to Group D failures shall be in accordance with procedures specified in MIL-STD-781.

4.5.5.3 Reinspection of Conforming Group D Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and sample units are subjected and pass Group A inspection.

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4.6 Preconditioning

4.6.1 Bounce Preconditioning - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. All equipment delivered on the contract or order shall be subjected to the requirements of this paragraph. If an equipment fails any test because of the bounce preconditioning, the defect applies to that test.

4.6.2 Electrical Preconditioning (Burn-in Debugging)

4.6.2.1 Sequence of Testing - Before subjecting equipments to this procedure, the contractor must have available objective evidence of compliance to Group A requirements.

4.6.2.2 Test Conditions - Every equipment shall be subjected to a 48 hour burn-in period. The last 24 hours of burn-in shall be failure free. The burn-in shall be conducted as if it were the beginning of a Production Reliability Test of Appendix 1 to this specification. The failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF.

4.6.2.3 Compliance - Units passing burn-in debugging shall be submitted to Group A, B and C inspection in that order without benefit of repair, cleaning realignment, or other changes, unless otherwise specified herein. If after successful completion of burn-in debugging, the equipment fails either the Group A or B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.7 Service Conditions - (Radio only: for antenna test procedures, see SM-A-696325 and for Mast, see SM-A-696070).

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4.7.1 Bounce Test; Class L Units - The equipment shall be tested in accordance with Method 514, Procedure IX, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII. During Part 1, all connections (such as cable assemblies) shall be in place. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.7.2 Shock Test; Ballistic - The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissable. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

4.7.3 Shock Test; bench-handling - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to and after testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

4.7.4 Temperature Test

4.7.4.1 Low Temperature - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -65°F and the operating temperature shall be -35°F. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.4.2 High Temperature - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 125°F. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1

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4.7.5 Humidity (moisture-resistance) test - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.6 Altitude (elevation) test - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810, except the chamber internal pressure in Step 2 shall be 282.0 mm of Hg (11.1 in. of Hg. or 25,000 ft. above sea level). During and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient atmospheric pressure for frequency changes.

4.8 Standard inspection conditions - Unless otherwise specified, all tests shall be performed at 115 volts ± 5 percent from a single-phase source operating between 47.5 and 63.0 HZ. Tests shall be conducted at room ambient temperature, pressure, and humidity unless otherwise specified.

4.9 Test specifications (service conditions) - The equipment shall be tested in accordance with the following test specifications and meet the requirements of this specification. The tests shall be performed during periods as specified in MIL-STD-810.

4.9.1 Temperature and humidity - 1.3 and 1.4 of SM-A-698051, except the test be performed at channels 390, 450, 600, 750, 900 and 1010 instead of those specified. Tests at line frequencies of 47.5 HZ and 420 HZ and high and low line voltage shall be performed at any one channel. 4.1 of SM-A-698049 with the exception that tests be performed at channels 390, 450, 600, 750, 900 and 1010 instead of those specified. 1.7 of SM-A-698051. 1.4 of SM-A-698050, except tests be performed at channels 400, 760 and 1010 instead of those specified. 1.2 of SM-A-698050, except test be performed at channel 600 instead of that specified. 4.2 of SM-A-698049, except frequencies below 10 KHZ be deleted from frequency response measurement. 4.5 of SM-A-698049. 4.6.8 of SM-A-698049. 4.7 of SM-A-698049.

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4.9.2 Altitude - 1.4 of SM-A-698051, except that the test be performed at channels 400 and 900 instead of those specified. 1.3 of SM-A-698051, except the test be performed at channel 900 instead of the channel specified. 1.7 of SM-A-698051, except that the test be performed at channel 1010 instead of the channels specified. 1.4 of SM-A-698050, except that the test be performed at channel 700 instead of the channels specified. 4.5 of SM-A-698049.

4.10 Operational Test Service Conditions - With the equipment set up in a system, an operational test shall be performed. The test shall consist of voice communications, order wire ringing through the systems plus a visual check of the metering facilities for reading specified in 4.1.6 and 4.2.1.2.6 of SM-A-698049.

4.11 A.C. Power Line, Voltage and Frequency Tests - The following tests shall be performed at room ambient and at channels 40, 180 and 400. The tests shall be performed at 105 volts 47 Hz, 115 volts 47 Hz, 125 volts 47 Hz, 105 volts 420 Hz, 115 volts 420 Hz and 125 volts 420 Hz.

Inspection	Requirement Paragraph
1.3 and 1.4 of SM-A-698051	3.4.1
1.4 of SM-A-698050	3.4.2.2
4.2.5 of SM-A-698826	3.4.3.7
4.2.2 of SM-A-698049	3.4.3.4
4.10 of this specification	3.4.4.1

4.12 Reliability Testing -

4.12.1 Reliability Qualification Test - The contractor shall perform a Reliability Qualification Test in accordance with Test Plan IV of MIL-STD-781 and Appendix I to this specification. The minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

4.12.2 Production Reliability Sampling - After successfully passing the Reliability Qualification Test, production reliability sampling shall begin. The Production Sampling Tests shall commence with the first lot available after the models used for first article testing. The samples shall be tested in accordance with Test Plan IVA of MIL-STD-781 and Appendix I to this specification. The specified (θ_0) MTBF shall be 1200 hours. The Minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

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4.12.3 Testing - All testing shall be conducted under the environment of Test Level B of MIL-STD-781. Equipment on-off cycling shall be on for 144 hours, off 8 hours. Vibration shall be for two minutes of every operating hour during one eight-hour period each week.

4.12.4 Failure Definition - A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractors approved test procedures. Paragraph 5.5 through 5.10 inclusive of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.4.8 of MIL-STD-781.

4.12.5 Rework and Retest Provisions - Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the lot unless positive action consisting of design change, modification, or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.13 Antenna Test Procedure - The antenna shall be tested in accordance with SM-A-696325 and meet the requirements of 3.4.5.

4.14 Mast Test Procedures - The mast shall be tested in accordance with SM-A-696070 and meet the requirements of 3.5.

4.15 Radio Set AN/GRC-103(V)2 Receiver Section Test Procedure - The receiver section shall be tested in accordance with paragraphs 1.3, 1.4, 1.5 and 1.6 of SM-A-698050 and meet the requirements of 3.4.2.1; 3.4.2.2; 3.4.2.3, 3.4.3.8 and 3.4.4.2; and 3.4.2.4 respectively.

4.16 Radio Set AN/GRC-103( )(V)2 Transmitter Section Test Procedure - The transmitter section shall be tested in accordance with paragraphs 1.3, 1.4, 1.5, 1.6 and 1.7 of SM-A-698051 and meet the requirements of 3.4.1.4, 3.4.1.1, 3.4.1.2, 3.4.1.3 and 3.4.1.5 respectively.

4.17 Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) Test - The receiver-transmitter, order wire shall be tested in accordance with 4.2.5 of SM-A-698807 and meet the requirement of 3.4.4.1.1.

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4.18 Radio Set AN/GRC-103( )(V)2 Test Procedure - The radio set shall be tested in accordance with paragraphs 4.2.3, 4.3, 4.4, 4.5, 4.6.7, 4.6.8, 4.7 and 4.8 of SM-A-698049 and meet the requirements of 3.4.3.3; 3.4.3.5; 3.4.3.6; 3.4.3.7.1 and 3.4.3.7.2; 3.4.4.8; 3.4.4.4; 3.4.2.6; and 3.8 respectively.

4.19 Component Test Procedures - The assemblies, subassemblies and modules of Radio Set AN/GRC-103( )(V) shall be tested in accordance with the following drawings in order to demonstrate compliance with 3.9 of this specification.

4.19.1 Receiver, Radio R-1329( )(P)/GRC-103(V) - The Receiver, Radio R-1329( )(P)/GRC-103 shall be tested in accordance with SM-A-698826 and meet the requirements thereof. Meeting paragraph 4.2.15 of this drawing demonstrates compliance with 3.4.2.5. Tests for 3.4.3.2 and 3.4.3.7.3 are found in paragraphs 4.2.12 and 4.2.8 of SM-A-698826 respectively.

4.19.2 Amplifier-Converter, AM-4317/GRC-103(V) - The Amplifier-Converter AM-4317( )/GRC-103(V) shall be tested in accordance with SM-A-698645 and meet the requirements thereof.

4.19.3 Transmitter, Radio T-983( )(P)/GRC-103 - The Transmitter, Radio T-983( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698831 and meet the requirements thereof. Tests for the requirements of paragraphs 3.4.4.3, 3.4.3.1, 3.4.3.4 and 3.4.1.7 are found in paragraphs 4.3.5, 4.9, 4.3.1 through 4.3.3 and 4.3.4 of SM-A-698831.

4.19.4 Amplifier-Frequency Multiplier AM-4321/GRC-103(V) - The Amplifier Frequency Multiplier AM-4321( )/GRC-103(V) shall be tested in accordance with SM-A-698649 and meet the requirements thereof.

4.19.5 Synthesizer (1RE1A2/5TR1A2) - The synthesizer shall be tested in accordance with SM-A-698628 and meet the requirements thereof.

4.19.6 Restorer, Pulse Form (1RE1A3) - The restorer, pulse form shall be tested in accordance with SM-A-698241 and meet the requirements thereof.

4.19.7 Limiter-Discriminator (1RE1A4) - The limiter-discriminator shall be tested in accordance with SM-A-698031 and meet the requirements thereof.

4.19.8 Panel Monitor (1RE1A1) - The panel, monitor shall be tested for continuity in accordance with SM-A-698317 and meet the requirements thereof.

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4.19.9 Power Supply (1RE1PS1) - The power supply shall be tested in accordance with SM-A-698372 and meet the requirements thereof.

4.19.10 Amplifier-Frequency Multiplier (1RE1A5) - The amplifier-frequency multiplier shall be tested in accordance with SM-A-698610 and meet the requirements thereof.

4.19.11 Amplifier, Video (1RE1AR1) - The amplifier, video shall be tested in accordance with SM-A-698035 and meet the requirements thereof.

4.19.12 Amplifier, Intermediate (1RE1AR2) - The amplifier, intermediate shall be tested in accordance with SM-A-698002 and meet the requirements thereof.

4.19.13 Distribution Box (1A2) - The distribution box shall be tested for continuity in accordance with SM-A-698685 and meet the requirements thereof.

4.19.14 Tuner, Radio Frequency (33A1) - The tuner shall be tested in accordance with SM-A-698645 and meet the requirements thereof.

4.19.15 Duplexer (33A1A1) - The Duplexer shall be tested in accordance with SM-A-698639 and meet the requirements thereof. Tests for the requirements of paragraphs 3.4.3.11, 3.4.3.9 and 3.4.3.10 are found in paragraphs 5.2; 5.3 and 5.4; and 5.5, 5.6 and 5.7 of SM-A-698639 respectively.

4.19.16 Converter, Frequency, Electronic (33A2) - The converter, frequency, electronic shall be tested in accordance with SM-A-698642 and the appropriate subassemblies shall be tested in accordance with SM-A-698640, SM-A-698641, and SM-A-698781 and meet the respective requirements thereof.

4.19.17 Mixer Stage, Frequency (33A7) - The mixer stage, frequency shall be tested in accordance with SM-A-696763-1 and meet the requirements thereof.

4.19.18 Control Indicator Coding Check (33A4) - The control indicator shall be tested in accordance with SM-A-696394 and meet the requirements thereof.

4.19.19 Control Monitor, Signal Level (33A5) - The control monitor shall be tested in accordance with SM-A-698643 and meet the requirements thereof.

4.19.20 Panel, Monitor (5TR1A1) - The panel monitor shall be tested in accordance with SM-A-698790 and meet the requirements thereof.

4.19.21 Control, Alarm (5TR1A3) - The control, alarm shall be tested in accordance with SM-A-698776 and meet the requirements thereof.

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4.19.22 Amplifier-Frequency Multiplier (5TR1A4) - The amplifier-frequency multiplier shall be tested in accordance with SM-A-696443 and meet the requirements thereof.

4.19.23 Amplifier-Monitor (5TR1A5) - The amplifier-monitor shall be tested in accordance with SM-A-698664 and meet the requirements thereof.

4.19.24 Power Supply (5TR1PS1) - The power supply shall be tested in accordance with SM-A-698363 and meet the requirements thereof.

4.19.25 Distribution Box (5A2A1) - The distribution box shall be tested in accordance with SM-A-698789 and meet the requirements thereof.

4.19.26 Amplifier, Radio (37AR1) - The amplifier, radio shall be tested in accordance with SM-A-698648 and meet the requirements thereof.

4.19.27 Frequency Multiplier Assembly (37A1) - The frequency multiplier assembly shall be tested in accordance with SM-A-698646 and meet the requirements thereof.

4.19.28 Control-Indicator (37A2) - The control-indicator shall be tested in accordance with SM-A-696394 and meet the requirements thereof.

4.19.29 Panel, Distribution (9A1) - The panel, distribution shall be tested in accordance with SM-A-698759 and meet the requirements thereof.

4.19.30 Panel, Monitor (9A2) - The panel, monitor shall be tested in accordance with SM-A-698695 and meet the requirements thereof.

4.19.31 Converter, Telephone Signal (9A3) - The converter, telephone signal shall be tested in accordance with SM-A-698805 and meet the requirements thereof.

4.19.32 Amplifier Assembly (9A4) - The amplifier assembly shall be tested in accordance with SM-A-698806 and meet the requirements thereof. Tests for the requirements of 3.4.4.11 and 3.4.4.5 are found in paragraphs 4.1.2, 4.1.5, 4.2.2, 4.3.2 and 4.3.5; and 4.2.5 of SM-A-698806.

4.20 Visual and Mechanical Inspection - Parts and equipment shall be examined for the defects listed in MIL-STD-252.

4.21 Mechanical Interchangeability - Mechanical interchangeability shall be demonstrated by gaging the applicable parts, assemblies and units with the gages listed on GL-SM-B-323442 and subordinate gage lists.

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4.22 Order Wire Operational Tests - Two terminals shall be set up to operate in a full duplex PCM system connected through sufficient attenuation. Frequency response and net loss measurements shall be made with the equipment meeting the requirements of 3.4.4.6 and 3.4.4.10.

4.23 Electromagnetic Interference Tests - The AN/GRC-103( )(V)2 shall be tested for compliance to the electromagnetic interference requirements of 3.4.1.6, 3.4.2.7 and 3.12. All test methods and setups shall be in accordance with the requirements of MIL-STD-462 and appendix of MIL-STD-461. Broadband omission tests shall be performed with the AN/GRC-103( )(V)2 operating at 550 MHz. All tests except CE06 shall be performed with the equipment operating in a full duplex mode.

4.24 Quality Conformance Inspection of Preparation for Delivery - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery provisions shall be in accordance with MIL-R-45292.

6. NOTES

6.1 Intended Use - The equipment covered by this specification is a complete, compact, rugged, radio relay set which may be used either as a terminal or modulation frequency patch repeater for intermediate or forward area systems. The radio set is used to provide simultaneous two-way (duplex) transmission. The system provides radio-frequency (rf) channel assignments from 395 MHz to 705 MHz. When used in combination, either as a repeater or as a terminal, it is possible to tune and operate the receiver and transmitter on independent frequencies over the frequency range.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation, packaging and packing (see Section 5).

c. When rough handling is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature, tools and running spare parts.

g. First Article inspection (see 3.1).

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6.3 Group C Inspection - Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.3).

6.3.1 Initial Shipments - Initial shipment of end items will be contingent only upon successful completion of all Group C tests (see 4.5.3).

6.4 Nomenclature - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-103W(V)2. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Color - The color chip furnished by the procuring agency will be color chip No. X24087 and may be obtained upon request to Commander, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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Project No. 5820-A619

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APPENDIX I

RELIABILITY TEST PROGRAM FOR RADIO SET AN/GRC-103(V)2

10. SCOPE

10.1 The Reliability Test Program for the Radio Set AN/GRC-103(V)2 shall be completed as per the following test plans:

a. Reliability Qualification Test, in accordance with MIL-STD-781, Test Plan IV, on four (4) systems to be selected from the First Article equipments.

b. Production Reliability Sampling Test, in accordance with MIL-STD-781, Test Plan IV A, on four (4) production systems selected from each production lot. A lot shall be defined for reliability test purposes as twenty-five (25) units.

10.2 The Reliability Demonstration shall be completed after reaching the acceptance criteria of plan IV or IVA (as applicable) for MIL-STD-781.

The MTBF (θo) for the above plans shall be 1200 hours for the system.

20. TEST CONDITIONS

20.1 The four equipments shall be set up in a two-way, two-hop system as shown in Figure 20.1 using suitable attenuators to absorb the output power from each transmitter and to provide an appropriate level of signal to each following receiver. The four equipments shall carry a P.C.M. signal and any deterioration in performance of any piece of equipment in the system shall produce error signals at the receiving end. The faulty equipment will then be located by checking individual meter readings.

20.2 The test shall be conducted in a suitable test chamber maintained at 40° ± 5°C. (95°F to 113°F).

20.3 The equipment shall be accessible to an operator for switching and retuning as required, and for interchanging of the R. F. Heads as called for in the test procedure.

20.4 The radio equipment shall operate continuously for periods of 144 hours and shall be switched off for periods of eight hours. During these "off periods" the R. F. Heads shall be interchanged as detailed in sub-section 30. A time recording meter shall be used to show total running time for the equipment.

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20.5 For two minutes of every operating hour during one eight-hour period each week, each equipment shall be vibrated at 2.2g ± 10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment.

20.6 The equipment undergoing reliability testing shall be subjected to a burn-in period of at least 48 hours prior to the start of reliability testing. This period of equipment operation shall be performed in accordance with applicable portions of the equipment specification.

30. TEST PROCEDURE

30.1 With the four radio sets connected as shown in Figure 20.1, set them to the first group of channels shown in Table 30.1.

30.2 With the P.C.M. test set on, adjust the transmitter No. A input control to provide ± 180 kHz peak deviation as indicated by the panel meter reading (in the green band when switched to 12 CH. P.C.M.).

30.3 Check the meter reading of the following receiver No. B when switched to 12 CH. P.C.M. It should show a reading in the green band.

30.4 Adjust the input control of transmitter No. C. for ± 180 kHz peak deviation as indicated on the panel meter and check the following receiver No. D.

30.5 Repeat for each equipment group in the "go" path and then back on the "return" path. Under normal conditions there shall be no appreciable error signal output from the P.C.M. test set and the electronic counter shall only give random indications. In the event of a fault developing anywhere in the system, error signals shall appear at the output and shall give visual indications on the counter. In order to provide continuous unattended monitoring, it is desirable that a digital recorder be connected to the output of the counter.

30.6 Every four hours throughout the test period, the operating channels shall be changed in accordance with the sequence in Table 30.1. Channel changes shall continue until an Accept/Reject decision is reach (see MIL-STD-781B, Tables IV and IVA).

30.7 Every six days the R.F. Tuning Heads shall be interchanged in accordance with the schedule in Table 30.2. This will be carried out during the off periods defined in sub-section 20. Tuning Heads shall be changed in the following manner:

a. The fixed heads shall be designated alphabetically and shall remain in fixed positions with the case. The R.F. Heads shall be designated numerically and shall be moved as required throughout the demonstration.

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b. The identifying numbers are prefixed by the letters T or R to indicate whether they are transmitting or receiving heads.

c. If a fourth (4) change is required, Table 30.2 will revert back to the start position and the sequence shall begin again; this shall continue as long as changes are required.

30.8 The meter readings of all equipments shall be logged, together with information on R.F. heads and channels used, after each frequency change. This log shall be in the form shown in Figure 30.1. Order Wire signalling shall be checked daily between all equipments on test.

40. FAILURE ACTION

40.1 Only failures occurring within the equipment under test directly as a result of normal operation shall be considered as relevant. Failures in the following categories are classed as non-relevant:

a. Failures in any of the test equipment used.

b. Failures due to accidents or improper test procedures.

c. Failures occurring during the burn-in period.

d. Failures which can be eliminated by mutually agreed corrections.

e. Failures caused by other failures.

40.2 When a suspected relevant failure occurs in any equipment, the test shall be stopped and the failed equipment removed, repaired and reinstalled at which time testing may continue.

40.3 The failed equipment shall be appropriately tagged to show the nature of the failure and the accumulated hours of operation. A tag shall be retained on the equipment and a notation of time of return to test added. Full details of the failure shall be recorded as detailed in sub-section 50 and the failure shall be recorded on the form shown in Figure 30.2.

50. TEST RECORDS

50.1 A complete operating log shall be kept during the test to record all pertinent information such as operating hours, off periods, vibration periods, channel changes, R.F. Head changes, meter readings and times at which failures occur or when units are restored to test.

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50.2 In addition to this operating log, an equipment failure record shall be kept for every equipment under test in the format given in specification MIL-STD-781.

50.3 To assist in the demonstration of M.T.B.F. for the equipment, a failure summary record shall be compiled as shown in MIL-STD-781 and shall include all failures classed as relevant on all equipments under test.

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TABLE 20.1

LIST OF TEST EQUIPMENT

DESCRIPTION	MANUFACTURER	TYPE
Vibration Machine 0-60 Hz	Lab Corporation	Model 30-300 or equivalent
Accelerometer	Columbia Labs	Model 1700 or equivalent
Random PCM Generator		USAECOM Dwg SM-E-698042 or equivalent
Electronic Counter	Hewlett-Packard	5245L or equivalent
Frequency Converter	Hewlett-Packard	5254C or equivalent
Directional Coupler	Hewlett-Packard	Model 778D or equivalent
Attenuator 60 dB, 2W	Hewlett-Packard	8491A-660 or equivalent

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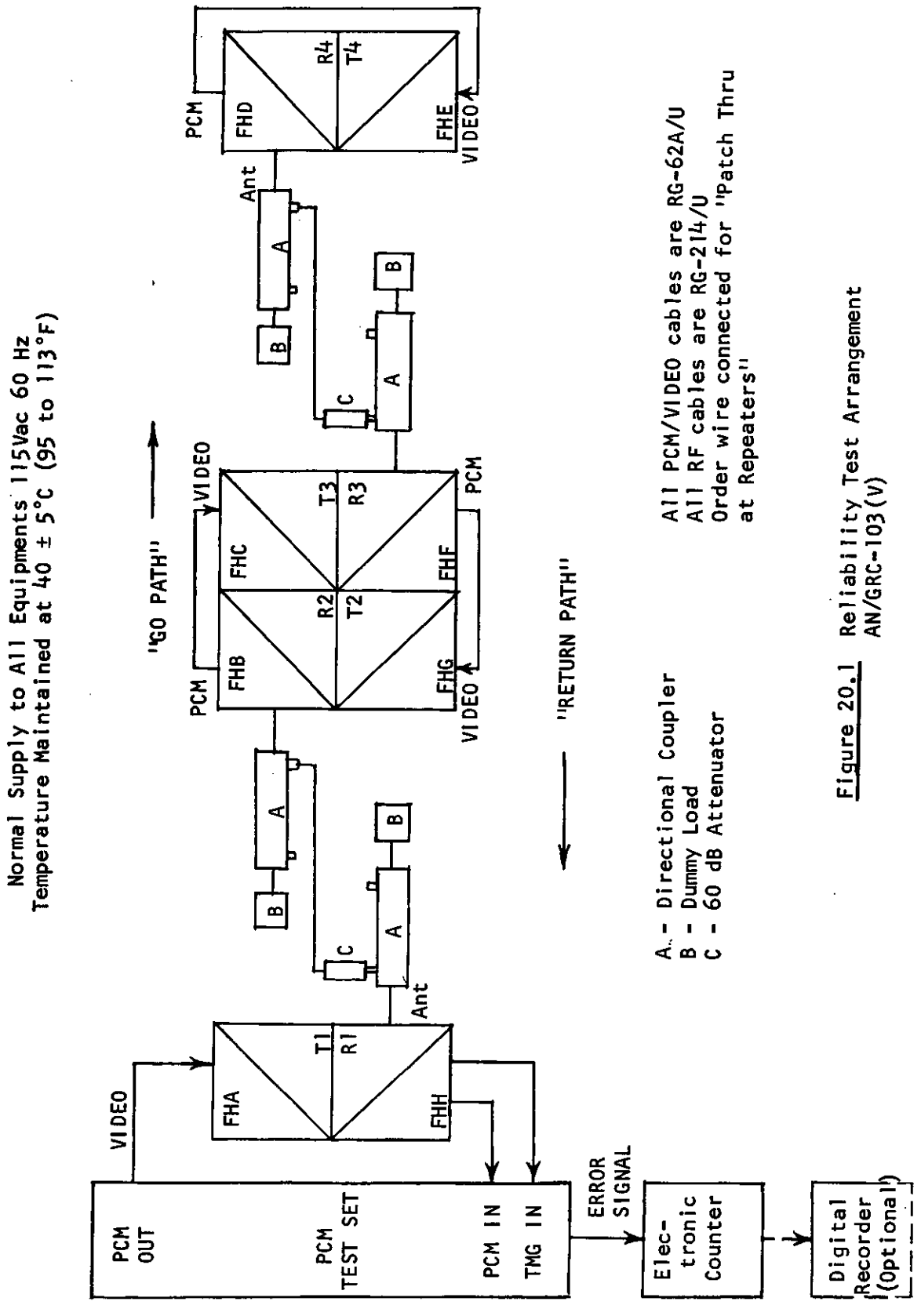


Figure 20.1 Reliability Test Arrangement AN/GRC-103(V)

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TABLE 30.1

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
0	390	596	390	596
4	802	1008	802	1008
8	391	597	391	597
12	803	1009	803	1009
16	392	598	392	598
20	804	1010	804	1010
24	393	599	393	599
28	805	394	805	394
32	600	806	600	806
36	395	601	395	601
40	807	396	807	396
44	602	808	602	808
48	397	603	397	603
52	809	398	809	398
56	604	810	604	810
60	399	605	399	605
64	811	400	811	400
68	606	812	606	812
72	401	607	401	607
76	813	402	813	402
80	608	814	608	814
84	403	609	403	609
88	815	404	815	404
92	610	816	610	816
96	405	611	405	611
100	817	406	817	406
104	612	818	612	818
108	407	613	407	613
112	819	408	819	408
116	614	820	614	820

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
120	409	615	409	615
124	821	410	821	410
128	616	822	616	822
132	411	617	411	617
136	823	412	823	412
140	618	824	618	824
144	413	619	413	619
148	825	414	825	414
152	620	826	620	826
156	415	621	415	621
160	827	416	827	416
164	622	828	622	828
168	417	623	417	623
172	829	418	829	418
176	624	830	624	830
180	419	625	419	625
184	831	420	831	420
188	626	832	626	832
192	421	627	421	627
196	833	422	833	422
200	628	834	628	834
204	423	629	423	629
208	835	424	835	424
212	630	836	630	836
216	425	631	425	631
220	837	426	837	426
224	632	838	632	838
228	427	633	427	633
232	839	428	839	428
236	634	840	634	840
240	429	635	429	635
244	841	430	841	430
248	636	842	636	842
252	431	637	431	637
256	843	432	843	432

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
260	638	844	638	844
264	433	639	433	639
268	845	434	845	434
272	640	846	640	846
276	435	641	435	641
280	847	436	847	436
284	642	848	642	848
288	437	643	437	643
292	849	438	849	438
296	644	850	644	850
300	439	645	439	645
304	851	440	851	440
308	646	852	646	852
312	441	647	441	647
316	853	442	853	442
320	648	854	648	854
324	443	649	443	649
328	855	444	855	444
332	650	856	650	856
336	445	651	445	651
340	857	446	857	446
344	652	858	652	858
348	447	653	447	653
352	859	448	859	448
356	654	860	654	860
360	449	655	449	655
364	861	450	861	450
368	656	862	656	862
372	451	657	451	657
376	863	452	863	452

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
380	658	864	658	864
384	453	659	453	659
388	865	454	865	454
392	660	866	660	866
396	455	661	455	661
400	867	456	867	456
404	662	868	662	868
408	457	663	457	663
412	869	458	869	458
416	664	870	664	870
420	459	665	459	665
424	871	460	871	460
428	666	872	666	872
432	461	667	461	667
436	873	462	873	462
440	668	874	668	874
444	463	669	463	669
448	875	464	875	464
452	670	876	670	876
456	465	671	465	671
460	877	466	877	466
464	672	878	672	878
468	467	673	467	673
472	879	468	879	468
476	674	880	674	880
480	469	675	469	675
484	881	470	881	470
488	676	882	676	882
492	471	677	471	677
496	883	472	883	472

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
500	678	884	678	884
504	473	679	473	679
508	885	474	885	474
512	680	886	680	886
516	475	681	475	681
520	887	476	887	476
524	682	888	682	888
528	477	683	477	683
532	889	478	889	478
536	684	890	684	890
540	479	685	479	685
544	891	480	891	480
548	686	892	686	892
552	481	687	481	687
556	893	482	893	482
560	688	894	688	894
564	483	689	483	689
568	895	484	895	484
572	690	896	690	896
576	485	691	485	691
580	897	486	897	486
584	692	898	692	898
588	487	693	487	693
592	899	488	899	488
596	694	900	694	900
600	489	695	489	695
604	901	490	901	490
608	696	902	696	902
612	491	697	491	697
616	903	492	903	492

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
620	698	904	698	904
624	493	699	493	699
628	905	494	905	494
632	700	906	700	906
636	495	701	495	701
640	907	496	907	496
644	702	908	702	908
648	497	703	497	703
652	909	498	909	498
656	704	910	704	910
660	499	705	499	705
664	911	500	911	500
668	706	912	706	912
672	501	707	501	707
676	913	502	913	502
680	708	914	708	914
684	503	709	503	709
688	915	504	915	504
692	710	916	710	916
696	505	711	505	711
700	917	506	917	506
704	712	918	712	918
708	507	713	507	713
712	919	508	919	508
716	714	920	714	920
720	509	715	509	715
724	921	510	921	510
728	716	922	716	922
732	511	717	511	717
736	923	512	923	512

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS			
	T1	T2	T3	T4
	R2	R1	R4	R3
740	718	924	718	924
744	513	719	513	719
748	925	514	925	514
752	720	926	720	926
756	515	721	515	721
760	927	516	927	516
764	722	928	722	928
768	517	723	517	723
772	929	518	929	518
776	724	930	724	930
780	519	725	519	725
784	931	520	931	520
788	726	932	726	932
792	521	727	521	727
796	933	522	933	522
800	728	934	728	934
804	523	729	523	729
808	935	524	935	524
812	730	936	730	936
816	525	731	525	731
820	937	526	937	526
824	732	938	732	938
828	527	733	527	733
832	939	528	939	528
836	734	940	734	940
840	529	735	529	735
844	941	530	941	530
848	736	942	736	942
852	531	737	531	737
856	943	532	943	532

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
860	738	944	738	944
864	533	739	533	739
868	945	534	945	534
872	740	946	740	946
876	535	741	535	741
880	947	536	947	536
884	742	948	742	948
888	537	743	537	743
892	949	538	949	538
896	744	950	744	950
900	539	745	539	745
904	951	540	951	540
908	746	952	746	952
912	541	747	541	747
916	953	542	953	542
920	748	954	748	954
924	543	749	543	749
928	955	544	955	544
932	750	956	750	956
936	545	751	545	751
940	957	546	957	546
944	752	958	752	958
948	547	753	547	753
952	959	548	959	548
956	754	960	754	960
960	549	755	549	755
964	961	550	961	550
968	756	962	756	962
972	551	757	551	757
976	963	552	963	552

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
980	758	964	758	964
984	553	759	553	759
988	965	554	965	554
992	760	966	760	966
996	555	761	555	761
1000	967	556	967	556
1004	762	968	762	968
1008	557	763	557	763
1012	969	558	969	558
1016	764	970	764	970
1020	559	765	559	765
1024	971	560	971	560
1028	766	972	766	972
1032	561	767	561	767
1036	973	562	973	562
1040	768	974	768	974
1044	563	769	563	769
1048	975	564	975	564
1052	770	976	770	976
1056	565	771	565	771
1060	977	566	977	566
1064	772	978	772	978
1068	567	773	567	773
1072	979	568	979	568
1076	774	980	774	980
1080	569	775	569	775
1084	981	570	981	570
1088	776	982	776	982
1092	571	777	571	777
1096	983	572	983	572

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
1100	778	984	778	984
1104	573	779	573	779
1108	985	574	985	574
1112	780	986	780	986
1116	575	781	575	781
1120	987	576	987	576
1124	782	988	782	988
1128	577	783	577	783
1132	989	578	989	578
1136	784	990	784	990
1140	579	785	579	785
1144	991	580	991	580
1148	786	992	786	992
1152	581	787	581	787
1156	993	582	993	582
1160	788	994	788	994
1164	583	789	583	789
1168	995	584	995	584
1172	790	996	790	996
1176	585	791	585	791
1180	997	586	997	586
1184	792	998	792	998
1188	587	793	587	793
1192	999	588	999	588
1196	794	1000	794	1000
1200	589	795	589	795
1204	1001	590	1001	590
1208	796	1002	796	1002
1212	591	797	591	797
1216	1003	592	1003	592
1220	798	1004	798	1004
1224	593	799	593	799

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RECEIVER METER READINGS (F.S.D.)

DATE	TIME	RF HEAD NO	CH NO	180 kHz 12CH	300 kHz 12CH	OW	+12V DC	-12V DC	REFL PWR	XMTR DUPL	OSC	DOUB	MULT	RCVR SIG	OW CHECK
LIMITS				40-60%	40-60%	40-60%	40-60%	40-60%	less than 15%	above 40%	20-90%	20-90%	above 25%	30-90%	Light on Buzzer on (1600 Hz)

TX FIXED HEAD

TRANSMITTER METER READINGS (F.S.D.)

DATE	TIME	RF HEAD NO	CH NO	180 kHz 12CH	300 kHz 24CH	12V DC	28V DC	600V DC	OSC	DOUB	MULT	DRIV	PWR OUT	REFL PWR	OW CHECK
LIMITS				40-60%	40-60%	40-60%	40-60%	40-60%	20-90%	20-90%	20-90%	20-90%	above 40%	less than 15%	Light on Buzzer on (1600 Hz)

FIGURE 30.1

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TABLE 30.2

SCHEDULE FOR INTERCHANGING OF R.F. TUNING HEADS

R.F. HEAD IDENT NBRS.	Fixed Head Identifying Letters				
	START	FIRST CHANGE	SECOND CHANGE	THIRD CHANGE	FOURTH CHANGE
T1	A	G	C	E	A
R2	B	F	D	H	B
T3	C	E	A	G	C
R4	D	H	B	F	D
T4	E	A	G	C	E
R3	F	D	H	B	F
T2	G	C	E	A	G
R1	H	B	F	D	H

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FIGURE 30.2 FAILURE DATA SHEET TECHNICIAN _____

PART 1 - TO BE COMPLETED AT TIME OF FAILURE

1. F.H./R.F. HEAD COMBINATION _____ /_____
2. DAYS/HOURS FROM START _____ / _____
3. CHANNEL AT TIME OF FAILURE CH. _____
4. SYMPTOMS OF FAILURE: _____

_____

_____

5. UNIT REMOVED FROM TEST (F.H. OR R.F. HEAD) _____
6. FAILURE REPORT FORM (F.R.F.) NO. _____
7. TOTAL DOWN-TIME OF FAILED UNIT _____

_____

PART 2 - TO BE COMPLETED WHEN F.R.F. IS CLOSED

1. SUB-UNIT FAILED _____
2. DWG. NO. _____
3. SERIAL NO. _____
4. COMPONENT REPLACED _____
5. COMMENTS: _____

_____

Signature _____

Date _____

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FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-49024(EL) RADIO SET AN/GRC-103(V)2

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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